

Honeywell

CL Server User's Guide

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About This Document

This document describes how to use the CL Server on the TPS Network from the server perspective. It does not cover the CL Application initiation from the TPN (LCN).

The primary user tasks are installation, configuration, and operation (startup and shutdown) of the CL Server. Additionally, you will access and view the System Management display for CL Server.

Release Information

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
System Management Operations Guide	EX07
System Management Configuration Guide	EX08
TPS System Administration Guide (for Windows 2000)	TP06X
TPS System Implementation Guide (for Windows 2000/XP)	TP08X
TPS System Planning Guide ((for Windows 2000/XP)	TP10X
System Configuration Utility User's Guide	TP20X
OPC Specification Reference Manual	TP41
HCI Client Developer Guide	TP60

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Contents

1. INTRODUCTION	9
1.1 Introduction to the CL Server	9
Purpose of this user guide.....	9
Summary.....	9
Features	9
1.2 Requirements for using the CL Server	10
User skill prerequisites	10
System environment	10
Server hardware and software environments.....	10
Network environments	11
How to use this guide	11
2. VERIFYING THE CL SERVER INSTALLATION.....	13
2.1 How to Verify a CL Server Installation	13
Who are the intended users for this section	13
What the CL Server package represents	13
Installed software packages that support CL Server	13
Other installation and uninstallation references.....	14
Verification of a successful installation	15
3. CONFIGURING THE CL SERVER.....	17
3.1 Approach to CL Server Configuration	17
About this section	17
3.2 CL Server Pre-configuration Requirements	18
Summary of pre-configuration requirements	18
Additional pre-configuration consideration	18
3.3 CL Server Configuration.....	19
Configuration overview.....	19
3.4 Invoking HCI Component Configuration Page.....	20
Procedure overview	20
Access HCI Component Configuration using System Management Display.....	20
Access HCI Component Configuration using Configuration Utility	22
3.5 Configuring CL Server from HCI Component Page.....	24
Procedure to configure CL Server	24
3.6 Selecting CL Server Security	27
Example Server Security Configuration page.....	27

3.7	CL Server Configuration Test.....	29
3.8	CL Server in a Workgroup	29
	Install package attempts to determine environment	29
	Account change and password resynchronization	29
3.9	CL Server Configuration Removal	30
	Procedure to remove component configuration.....	30
4.	MONITORING CL SERVER STATUS	31
4.1	Overview of CL Server Status Operations	31
	CL Server appears in the System Management Display	31
	Typical user interface operations.....	31
4.2	Interpreting System Status Indicators.....	32
	Status information in both the Scope and Results pane.....	32
	Scope pane	33
	Results pane	34
	Component status	35
	Synchronization between Nodes.....	35
4.3	Starting the System Management Display	36
	Two methods to start the System Management Display	36
	Invoke the System Management Display from the Start menu.....	36
	Invoke the System Management Display from the Run command	37
4.4	Starting the CL Server	38
	Start the CL Server.....	38
	Confirmation of CL Server Startup	40
	CL Server is Started and in Idle Mode.....	41
4.5	Invoking the CL Server Auxiliary Status Display	42
	Invoke the Auxiliary Display	42
	View the CL Server Status Display.....	43
4.6	Shutting Down the CL Server	44
	Begin the CL Server shutdown.....	44
	Shut down the CL Server	45
4.7	Auxiliary Status Display – CL Server.....	46
	CL Server Auxiliary Status Display organization	46
	General Data section.....	47
	Security Data section	48
	CL-Initiated Application Data section.....	48

5.	SUPPORTING CL APPLICATIONS.....	49
5.1	Application Directory Summary	49
General		49
Using shortcuts		49
Precedence in launching CL Initiated applications		49
Accessing data for CL-Initiated application		50
5.2	CL Specific Language Support.....	51
Supporting Functions		51
Initiate_Task.....		51
Activate_Task.....		55
Terminate_Task		56
Execute_Task_With_Wait		57

Contents

1. Introduction

1.1 Introduction to the CL Server

Purpose of this user guide

This guide describes how to use the CL Server on the TPS Network (TPN) from the server perspective. It does not cover the CL Application initiation from the TPN (LCN).

The primary user tasks are installation, configuration, and operation (startup and shutdown) of the CL Server. Additionally, you will access and view the System Management display for CL Server.



ATTENTION

This document assumes that you have already installed System Management components. Refer to the System Management Configuration Guide if you need to install the System Management Display.

Summary

The CL Server accepts requests from the TPN to initiate, run, and terminate Windows 2003-side applications. It retains all the functionality of previous CL controllers. Any CL-initiated applications that ran on an Application Module Personality (AMW) will run on a TPS system.

The CL Server is an HCI managed component, and can be viewed and manipulated from the System Status Display.

A windowless application on the Windows 2003-side can be started and stopped by a CL program on the TPN side at anytime. CL Initiated Applications are also managed using the System Management Display. The CL Server component Status Display will show the status of the CL Server and initiated applications.

Features

The CL Server User Guide provides information for you to do the following:

- Verify the CL Server installation.
- Configure the CL Server.
- Operate the CL Server (startup, viewing status displays, and shutdown).

1. Introduction

1.2. Requirements for using the CL Server

1.2 Requirements for using the CL Server

User skill prerequisites

This document is a guide for navigating and accessing information that will allow the CL Server to become functional. It is assumed that the user is skilled in performing Windows administration tasks and has a working knowledge of the TPS system.



ATTENTION

CL Server implementation requires that you are logged on as both a local administrator to install server component software. When in a domain environment, to configure the CL Server you must be logged on as a TPS Administrator as to configure the TPS component and configure security. Therefore, you must be familiar with the system policies contained in the *TPS System Administration Guide* and *TPS System Implementation Guide*.

System environment

The CL Server may run on a Windows 2003 server optionally configured as part of the TPS domain. This workstation must be directly connected to the TPN through an LCNP board in the PC. The network requires the configuration for use of DCOM.

Server hardware and software environments

Refer to the Software Change Notice that accompanies the TPN Server package as well as the TPS System Implementation Guide for hardware and software requirements.

Network environments

Item	Requirement
Network	• If using an optional TPS domain, then the network requires the availability of a Windows Domain Controller (Windows 2000 Server or Windows Server 2003). • If using workgroups, domain functions such as a TPS Domain cannot be introduced.
PIN/PCN	Ethernet, TCP/IP, Microsoft application layer software such as RPC & DCOM. One of the following name services must be available: • DNS – Required for Windows domains • WINS – Optional and recommended.

How to use this guide

This guide is organized into the following major sections that represent user tasks:

The *Verifying the CL Server Installation* section provides a method of verifying the installation of the CL Server component software. This section may be skipped if the CL Server is already installed and functional.

The *Configuring the CL Server* section describes how to configure the CL Server as a component in an optional TPS Domain.

The *Monitoring CL Server Status* section provides operational procedures that allow you to view status, start, and stop the CL Server.

The *Supporting CL Application's* section provides information about CL-initiated Windows applications.

1. Introduction

1.2. Requirements for using the CL Server

2. Verifying the CL Server Installation

2.1 How to Verify a CL Server Installation

Who are the intended users for this section

This section is of interest primarily to the system administrators or to anyone who needs to verify the installation of the CL Server software package on their system. If CL Server is already installed and functioning on your system, you may skip this section.



ATTENTION

The intent of this section is to provide a method of verifying the installation of the CL Server software. The *TPS System Implementation Guide* provides the complete instructions for installing all of the TPS software. This includes the CL Server software package according to the licensing agreement. Please consult that publication for installation-related tasks.

What the CL Server package represents

The CL Server is a specific package component within the TPS software package.

The CL Server software package must be installed on a TPS node that contains an installed LCNP board. Currently, there can be only one CL Server for each TPS APP Node.

Installed software packages that support CL Server

The CL Server software installation can be verified assuming the following software packages are installed:

1. TPS Base Software
2. Emulators
3. HCI Runtime
4. System Management Runtime
5. System Management Display (Optional)

2. Verifying the CL Server Installation

2.1. How to Verify a CL Server Installation

Other installation and uninstallation references

The TPN Server software package and other TPS packages can be uninstalled. The uninstall procedure is described in the Software Change Notice that accompanies the CL product.



ATTENTION

Verify that the CL Server software **is installed on an APP Node**. The node that is assigned to run the CL Server must contain an LCNP board. Currently, there can be only one CL Server per LCNP board.



REFERENCE - EXTERNAL

Please reference the *TPS System Implementation Guide* for complete details on necessary prerequisites and installing the TPS software

2. Verifying the CL Server Installation

2.1. How to Verify a CL Server Installation

Verification of a successful installation

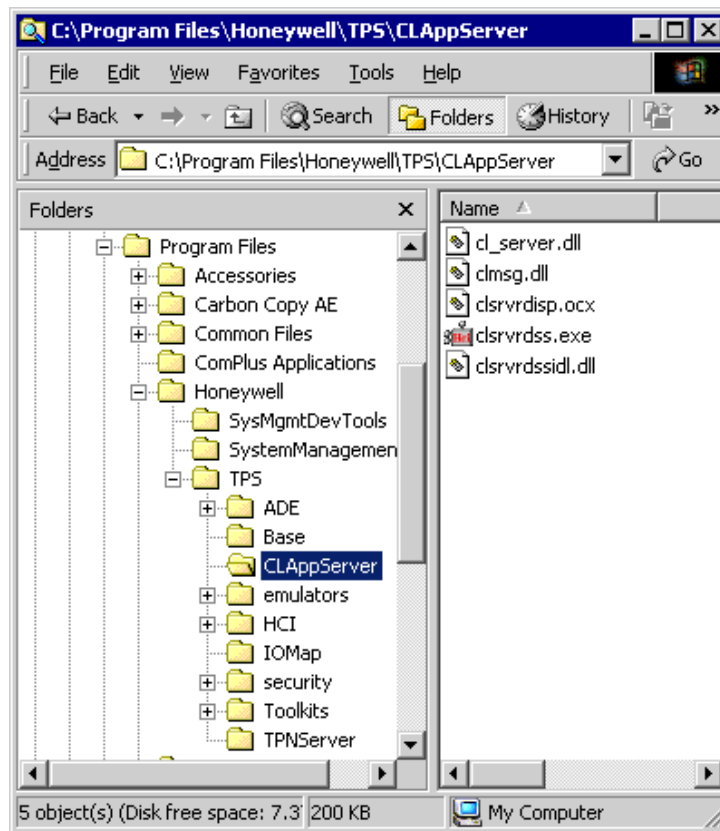
Upon successful CL Server installation, your selected drive directory should include a folder called CLAppServer containing the contents similar to the items shown in the following figure.



ATTENTION

The contents of the CLAppServer folder will vary depending on what kind of node is being used and what software packages are being installed

Figure 1 Directory with the CLAppServer Folder



2. Verifying the CL Server Installation

2.1. How to Verify a CL Server Installation

3. Configuring the CL Server

3.1 Approach to CL Server Configuration

About this section

This section provides the following information about CL Server configuration.

- CL Server pre-configuration requirements
- CL Server configuration
- CL Server installation test
- CL Server configuration removal

3.2 CL Server Pre-configuration Requirements

Summary of pre-configuration requirements

The following table lists CL Server pre-configuration requirements.

Pre-configuration Requirement	Note the Following	For Additional Information
Configure the Application Processing Platform (APP) node	Be sure to select the CL/NT Application Server when installing TPS Software. The <i>CL/NT Application Server</i> should be installed on the APP Node where the component will run.	TPS System Implementation Guide
Install the connectivity client on nodes that access the CL Server	The <i>CL/NT Server Connectivity Client</i> should be installed on nodes from which the component will be configured or monitored. Note that the <i>CL/NT Server Connectivity Client</i> item is automatically selected when the GUS HCI Client Add-In package is selected.	TPS System Implementation Guide

Additional pre-configuration consideration

The node hosting the CL Server establishes its HCI Name Service scope based on the hosting node's environment.

- If the node is in a workgroup, its HCI Name Service scope is the workgroup.
- If the node is in a domain, its HCI Name Service scope is the domain.
- If the node is in a TPS Domain, its HCI Name Service scope is the TPS Domain.

The HCI Name Service is used by HCI client applications to resolve an alias name to a server's CLSID and computer name. The alias name for CL Server must be unique within its scope.



REFERENCE - INTERNAL

For more information about System Management facilities such as HCI Name Service, refer to the [System Management Configuration Guide](#).

3.3 CL Server Configuration

Configuration overview

CL Server configuration consists of defining the server as an HCI managed component. The following table provides an overview of how to configure a CL Server.

Step	Action	Result
1	Verify the pre-configuration requirements listed previously are in place.	APP node and client nodes have base components installed.
2	Access the HCI Component Configuration page using either the System Management Display or Configuration Utility.	The HCI Component Configuration page provides definable CL server properties.
3	Configure the CL Server using the component properties to define the following: • Select HCI CL Server as the BASE Prog ID property. • Name the CL Server (for example: CL Server1) using the Component Name property. • Determine security (for example: access levels) using the Secured Methods and Capability properties.	A CL Server is configured and ready to support client application access.

3. Configuring the CL Server

3.4. Invoking HCI Component Configuration Page

3.4 Invoking HCI Component Configuration Page

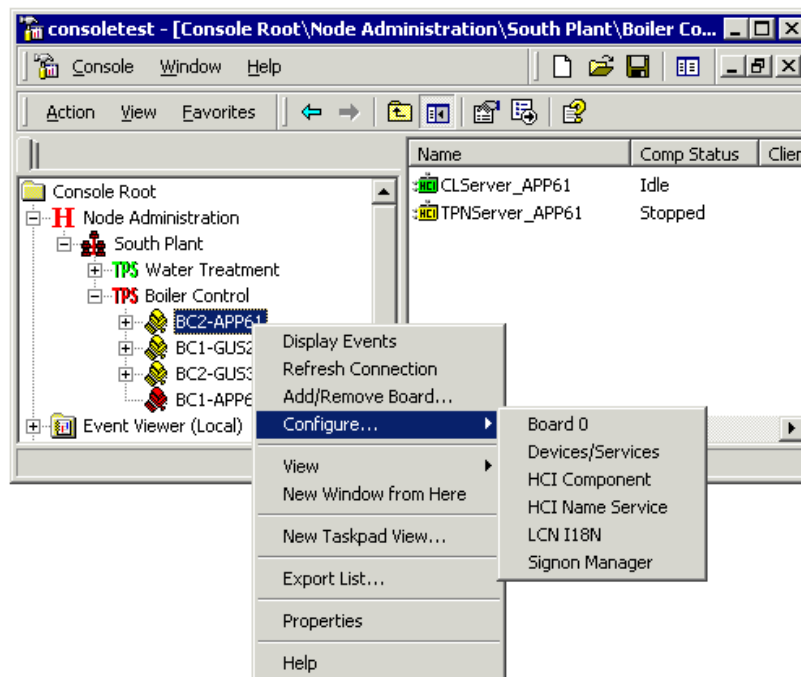
Procedure overview

HCI Component Configuration operates within the System Management Display that is implemented as a Microsoft Management Console (MMC) snap-in. Alternatively; the HCI Component page is also accessible from the Configuration Utility. To configure a CL Server, first access the HCI Component Configuration page using one of the following procedures.

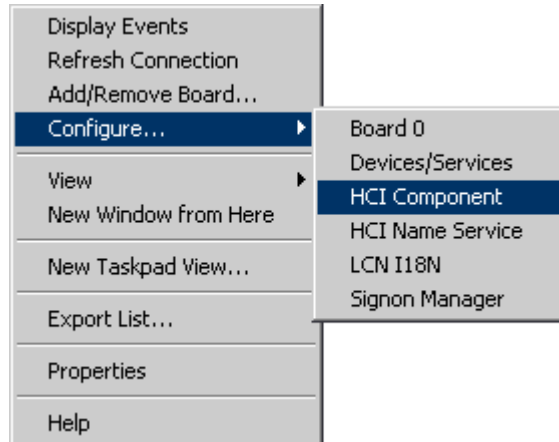
Access HCI Component Configuration using System Management Display

Step	Action
------	--------

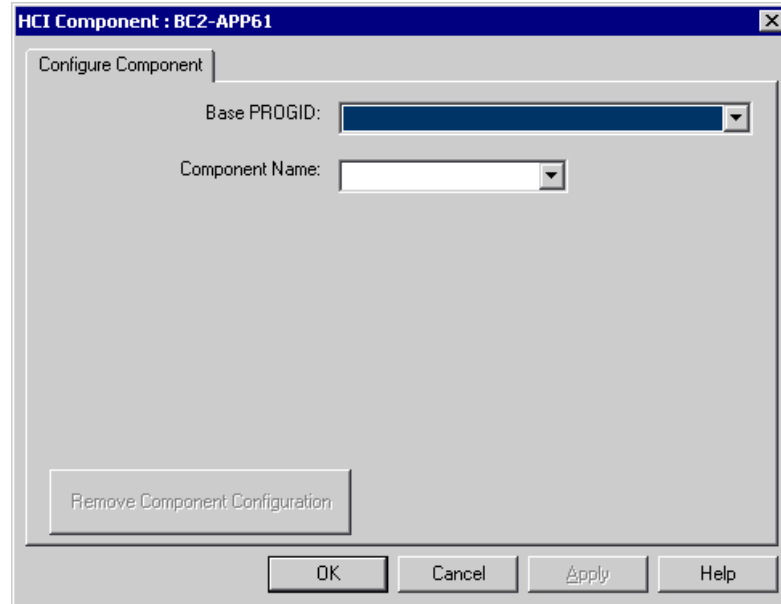
- | | |
|---|--|
| 1 | Right-click the computer item representing the APP node from the System Management Display hierarchy and select Configure . |
|---|--|



Step	Action
2	Select HCI Component from the menu to begin CL Server configuration.



Result: The HCI Component dialog appears



3. Configuring the CL Server

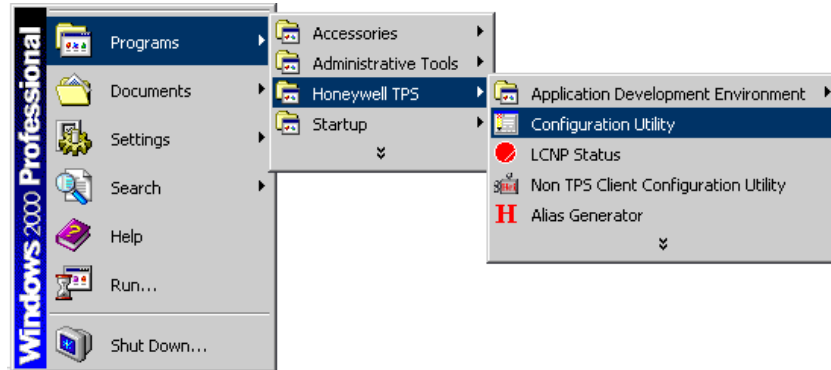
3.4. Invoking HCI Component Configuration Page

Access HCI Component Configuration using Configuration Utility

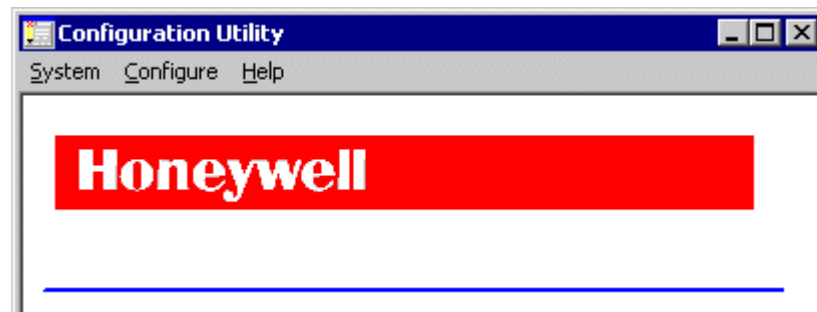
Step

Action

- 1 Select **Program>Honeywell TPS>Configuration Utility**



Result: The Configuration Utility appears.

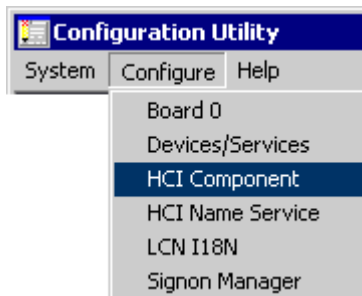


3. Configuring the CL Server

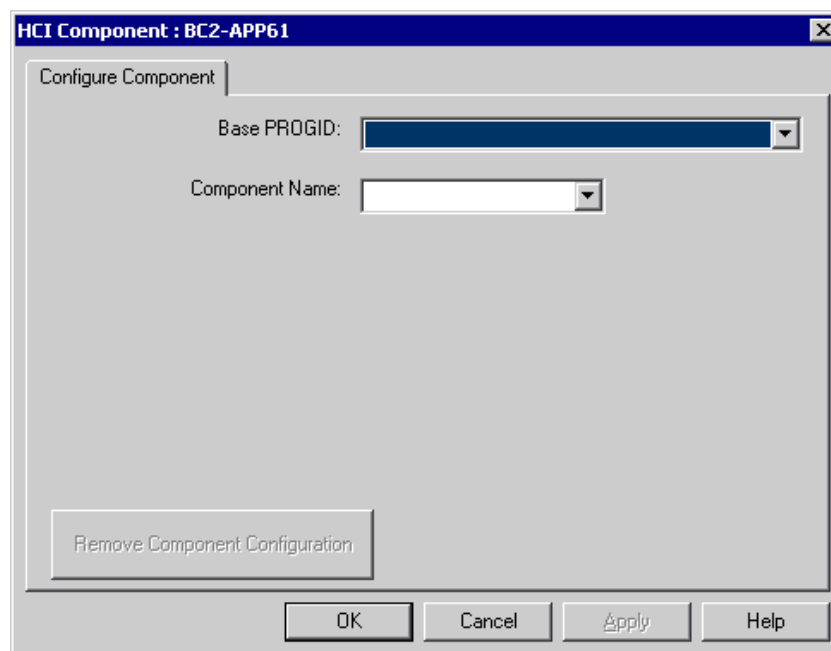
3.4. Invoking HCI Component Configuration Page

Step	Action
------	--------

2	Select Configure>HCI Component .
---	--



Result: The HCI Component page appears.



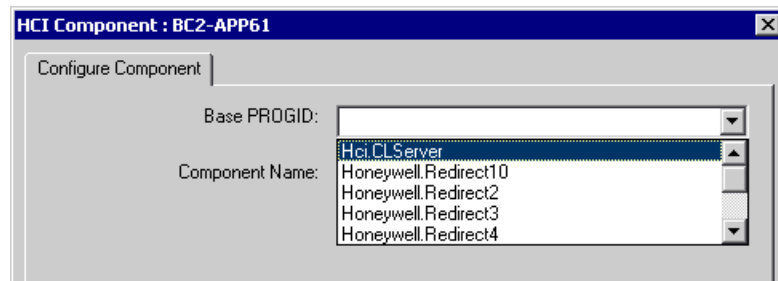
3. Configuring the CL Server

3.5. Configuring CL Server from HCI Component Page

3.5 Configuring CL Server from HCI Component Page

Procedure to configure CL Server

Step	Action
1	Select Hci.CLServer from the Base PROGID drop-down list. Result: The Configuration Page displays definable CL Server properties.



3. Configuring the CL Server

3.5. Configuring CL Server from HCI Component Page

Step

Action

- 2 Enter a name in the **Component Name** field for the CL Server that will be meaningful to your plant's users

The screenshot shows a Windows-style dialog box titled "HCI Component : BC2-APP61". It contains several configuration fields and buttons. The "Base PROGID:" field is a dropdown menu showing "Hci.CLServer". The "Component Name:" field is an empty text box. There is a checkbox for "Local Component" which is unchecked. The "Device ID:" field is an empty text box. The "AutoStart:" field has two radio buttons, "YES" and "NO", with "NO" selected. The "Auxiliary Status Display ProgID:" field contains the text "CLSRVRDISP.ClsrvrdispCtrl.1". Below this is a table for "Secured Methods:" with two columns: "Method" and "Capability". The table has one row with "Shutdown" in the Method column and "<OPEN ACCESS>" in the Capability column. Below the table is an "Edit Capability:" dropdown menu. At the bottom of the dialog are two buttons: "Remove Component Configuration" and "Enter/Edit Server Specific Configuration...". At the very bottom are four buttons: "OK", "Cancel", "Apply", and "Help".

Method	Capability
Shutdown	<OPEN ACCESS>

3. Configuring the CL Server

3.5. Configuring CL Server from HCI Component Page

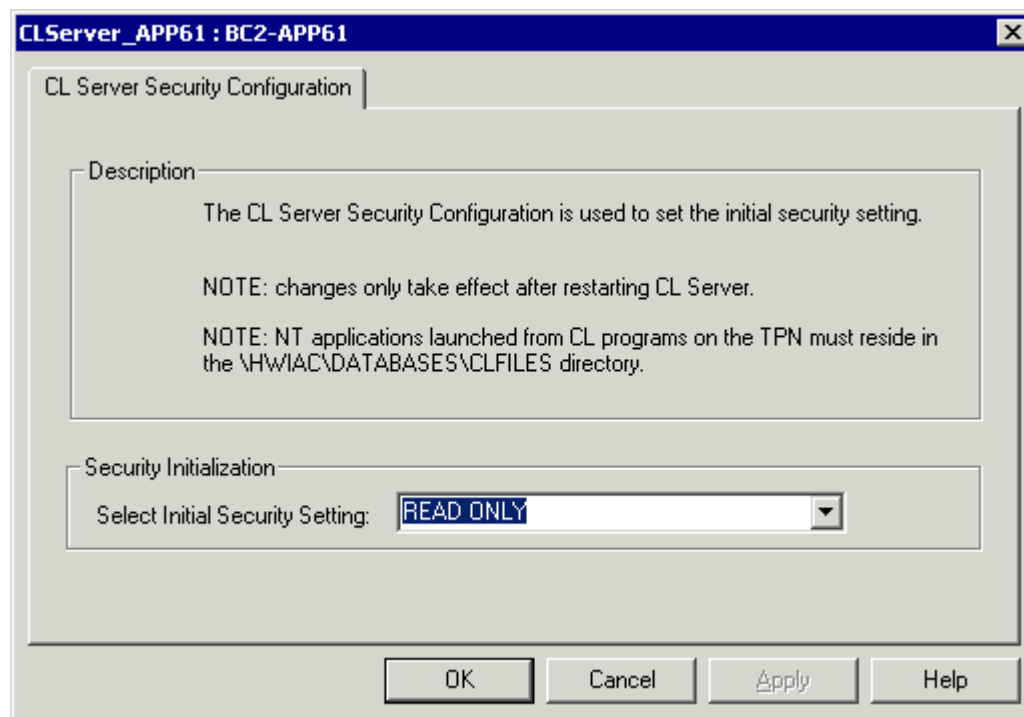
Step	Action
3	Click the Check Name button to verify the validity of the Component Name. 
4	Select the NO AutoStart option if the CL Server is to be manually started Or, Select the YES AutoStart option if the CL Server is to be started at every system restart.
5	Click the Enter/Edit Server Specific Configuration button.
6	Click the OK button for each message box that appears until just the CL Server Security Configuration page is displayed and then continue to the <i>Selecting CL Server Security</i> section.

3.6 Selecting CL Server Security

Example Server Security Configuration page

An example of the **CL Server Security Configuration** page is shown in the following figure:

Figure 2 CL Server Security Configuration Page



The global X-access switch controls when a Windows 2003-side application can write TPS Network data. The **Initial Security Setting** shown on the **CL Server Security Configuration** page provides the startup value for the global X-access switch.

It should be noted that this switch is only used when the CL Server is running. If the CL Server is not running, all Windows 2003-side applications will be able to write TPS Network data. The switch is implemented by a Processor Status Data Point (PSDP) parameter called \$XACCESS which has three possible states:

3. Configuring the CL Server

3.6. Selecting CL Server Security

- READ ONLY-The Windows 2003-side can read but cannot write TPS Network data (default).
- READ WRITE ONLY FOR CL INITIATED-The Windows 2003-side can read TPS Network data, but can write TPS Network data only from applications that are initiated by CL on the TPS Network-side.
- READ WRITE-The Windows 2003-side can read and write TPS Network data from CL-initiated and non-CL-initiated applications (this state cannot be set by any means unless the external load module XACCESS is loaded).



ATTENTION

For nodes that run CL Server and TPN Server the **Initial Security Setting (\$XACCESS)** must be set to READ-WRITE. Currently, \$XACCESS security is limited when Windows 2003 Applications use the TPN Server to access TPN data. The TPN Server only supports read-only access and read-write access. When read-write only for CL Initiated Applications is selected, the TPN Server will default to read-only access.

The setting configuration is described in the following procedure.

Step	Action
1	Select a security setting from the Select Initial Security Setting pull-down menu using the preceding security setting description information.
2	Click OK to re-display the HCI component Configuration page.

3.7 CL Server Configuration Test

Step	Action
1	Access the System Management Display. Example: Program > Administrative Tools > [your console].msc
2	If the AutoStart option for the CL Server is selected, then the System Management Display should show that the CL Server has started Or, If the AutoStart option is not selected, then start the server by right-clicking the server name in the System Management Display and then click All Tasks > Start . Click Yes to confirm the start.

3.8 CL Server in a Workgroup

Install package attempts to determine environment

During installation of CL Server, the install package attempts to determine if CL Server is being installed in a domain or workgroup environment. If installed in a domain, CL Server uses a TPSApp domain account. If installed in a workgroup, CL Server uses a TPSLocalServer local account. The respective accounts are available from the TPS workstation security template.

Account change and password resynchronization

If it is necessary to change the DCOM configuration to use a local account from a domain account, you can change accounts using dcomcnfg (Go to Start->Run->dcomcnfg. Select Hci.CLServer. From the Identity tab, change the user to "TPSLocalServer". Enter a password for this account.) If you do not know the accounts' password you will have to change it to something you know. If you change the password, you must change it on all nodes in the workgroup. After changing the password, you will need to resynchronize all Windows Services and DCOM servers that use the TPSLocalServer account to the new password. This also must be done on all nodes in the workgroup using the password utility, **Windows Services & DCOM Servers Log on tool**.

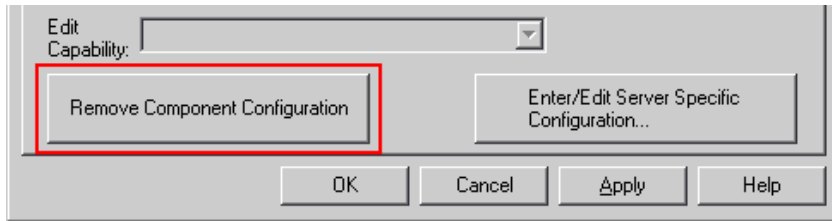


REFERENCE - INTERNAL

Refer to "Password Security and Synchronization" in the *TPS System Administration Guide* for procedures on how to use the password utility.

3.9 CL Server Configuration Removal

Procedure to remove component configuration

Step	Action
1	Right-click the node that contains the CL Server from the System Management Display.
2	Select Configure>HCI Component from the Context menu. Result: The HCI Component Configuration Page appears.
3	Select CL Server from the Component Name list to display the Configure Component page.
4	Click the Remove Component Configuration button. 
5	Access the HCI Name Service to verify the name is removed from the HCI Name Service Repository. a) Right-click the node that contains the CL Server. b) Select Configure>HCI Name Service .
6	From the HCI Name Service page, select Refresh List and verify the name is removed.

4. Monitoring CL Server Status

4.1 Overview of CL Server Status Operations

CL Server appears in the System Management Display

Because the CL Server is a managed component in the TPS system, it appears in the System Management Display. The CL Server provides status information to both the scope pane and the results pane of the System Management Display.

The managed CL Server can be configured to be auto-started upon system power-up by the Component Administration Service (CAS) provider. Alternatively, a managed component may be manually started from the System Management Display or from a client with appropriate access levels.

This section describes how you can manually start up and shut down a CL Server through the System Management Display.

Typical user interface operations

From the System Management Display, you can:

- Start up the CL Server
- View the status of HCI Components including the CL Server
- Invoke the CL Server Auxiliary Status Display
- Shut down the CL Server

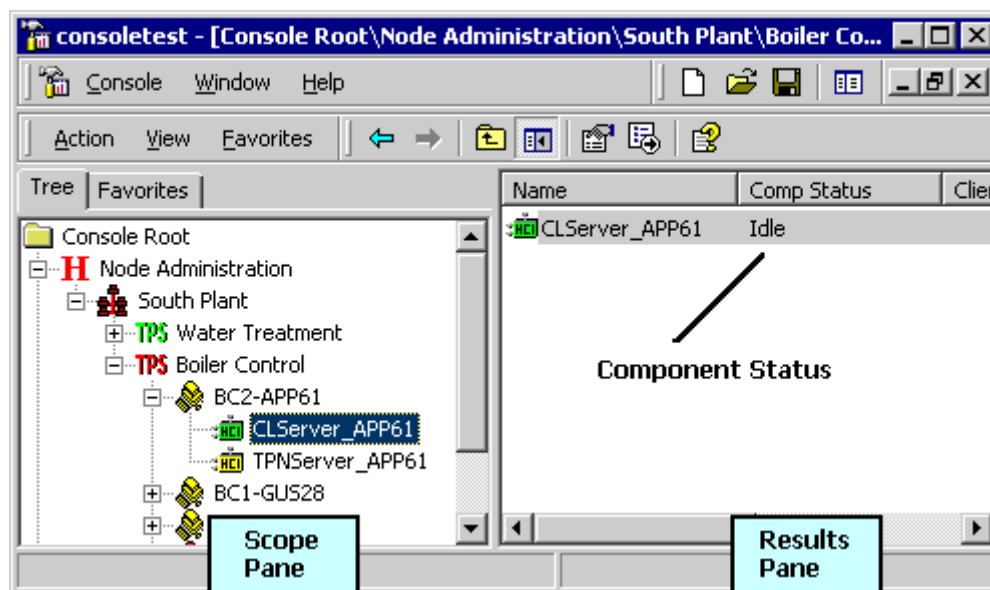
The information in the following procedure is basic reference material for the System Management Display. The procedure provides a description of the display and defines the entities used within the status display window.

4.2 Interpreting System Status Indicators

Status information in both the Scope and Results pane

The following shows the System Management Display with the scope and results pane. The scope pane shows the Node Administration snap-in as the root of the tree and each of the static computers, domains, or TPS Domains selected for monitoring. The results pane shows the status of an HCI managed component.

Figure 3 System Management Display



Scope pane

The scope pane shows a tree display composed of several levels. Double-click a level to expand the tree display until you can view the HCI Managed Component that is of interest.

The scope pane (left pane) provides an icon that gives you a quick visual indication of the state of the component. The icon can be one of three colors as shown in the table below:

Component State Color (Scope Pane Icon)	Component State Value
Green	Component is in the Running or Idle State. The TPS Network Personality is running and the server is fully functioning.
Yellow	Component is in the Warning or Stopped state. A) It is waiting for the TPS Network Personality to be loaded and transition to the running state on the LCNP in the local node. B) It has lost its connection to the TPS Network Personality.
Red	Component is <u>not</u> in Running or Warning State. The System Management Display will default to the Stopped state if the CL Server is not running.

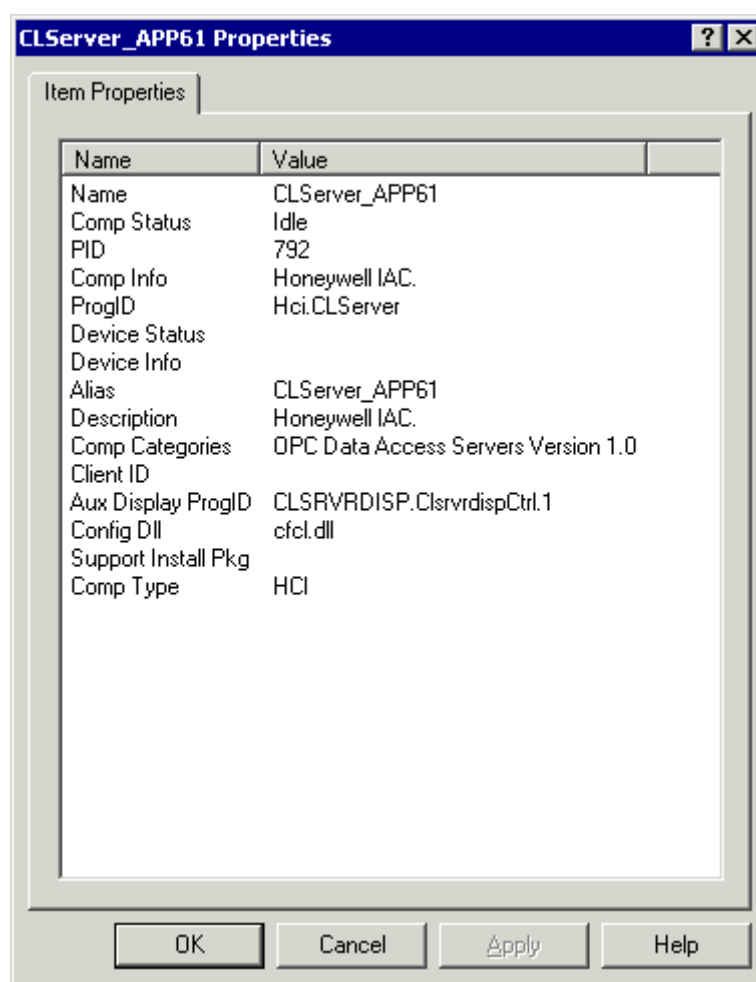
4. Monitoring CL Server Status

4.2. Interpreting System Status Indicators

Results pane

The results pane (right pane) presents the status for the selected item from the scope pane. The results pane presents information in configurable columns. The following figure lists properties that can appear in the results pane.

Figure 4 CL Server Properties



To change column width for any of these columns, drag the column separator to the left or right. To sort a column, click on the column header.

The Information column field is a string of 40 characters or less. The CL Server will only provide useful (non-NULL string) information for this field if it is in the Warning State.

Component status

The Results pane also provides a Component Status column that contains the status of the component. The possible status values are:

Components Status	Icon Color	Server Status Values
Stopped	YEL	Component is not currently operational, but has been configured on the node.
Initializing	YEL	Component is initializing.
Test	YEL	Component is in a testing state.
Idle	GRN	Component is operational, and no CL Initiated Applications are running.
Running	GRN	Component is operational, and CL Initiated Applications are running.
Warning	YEL	Component is running, but it has a problem that is not disabling the component.
Shutting down	YEL	Component is shutting down.
Failed	RED	Component terminated abnormally.

Synchronization between Nodes

The System Management Displays on the TPS Network update independently of each other. This means that displays on different nodes may show a different status between configurable update cycles (typically every 15 seconds).

4. Monitoring CL Server Status

4.3. Starting the System Management Display

4.3 Starting the System Management Display

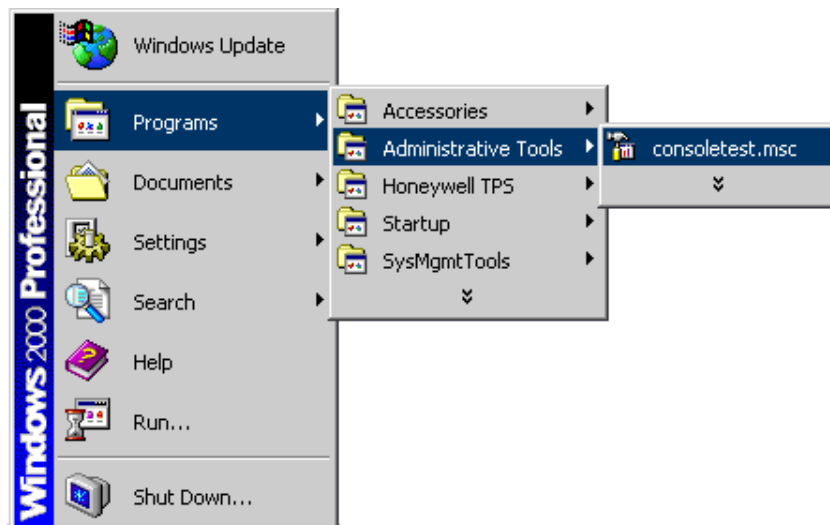
Two methods to start the System Management Display

Use one of two methods to start the System Management Display.

- Invoke using the Start menu
- Invoke from the Run command

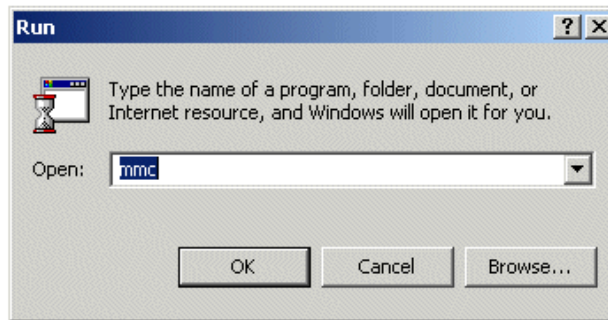
Invoke the System Management Display from the Start menu

Step	Action
1	Invoke the System Management Display by performing the following: Click Start>Programs>Administrative Tools
2	Click [console].msc .



Invoke the System Management Display from the Run command

- | Step | Action |
|------|---|
| 1 | Invoke the System Management Display by using the Run command.
Type mmc in the Run dialog. |



- | | |
|---|--|
| 2 | Select a previously saved console view when the MMC appears. |
|---|--|
-

4. Monitoring CL Server Status

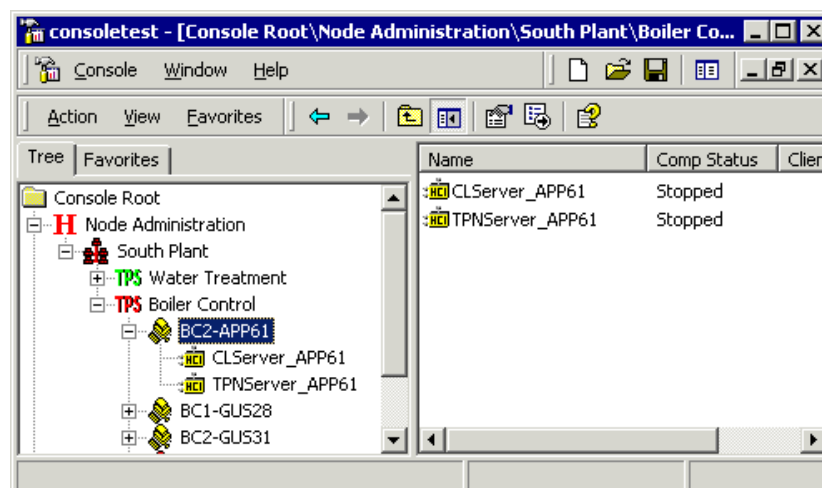
4.4. Starting the CL Server

4.4 Starting the CL Server

Start the CL Server

Step	Action
1	Verify the TPS node personality is loaded and running on the LCNP board before starting the CL Server.
2	Invoke the System Management Display using one of the methods in the procedure <i>Starting the Status Management Display</i> .

Result: The **System Management Display** appears showing icons (in tree form) in the scope pane (left side) and HCI component information in the results pane (right side).



4. Monitoring CL Server Status

4.4. Starting the CL Server

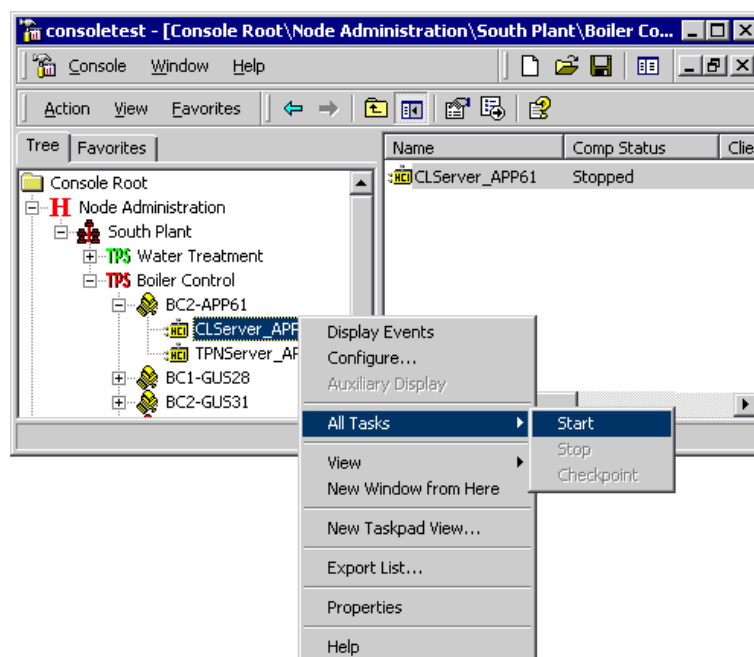
Step	Action
3	Select or click the node icon where the CL Server is configured (this example uses CLServer_APP61). Example: The State of the HCI component CLServer_APP61 is indicated as Stopped. Note: Visual State of Component (icon color) Green: Running or Idle state Yellow: Warning state (example) – Waiting for TPS Node Personality to load – Lost connection to TPS Node Personality Red: Not in Running or Warning state Component Status Idle: Server is operational, and no CL-Initiated Applications are running Running: Server is operational, and CL-Initiated Applications are running Stopped: Server is configured, but not running. Warning: TPS Node personality is not loaded in LCNP or connection to TPN has been lost.

4. Monitoring CL Server Status

4.4. Starting the CL Server

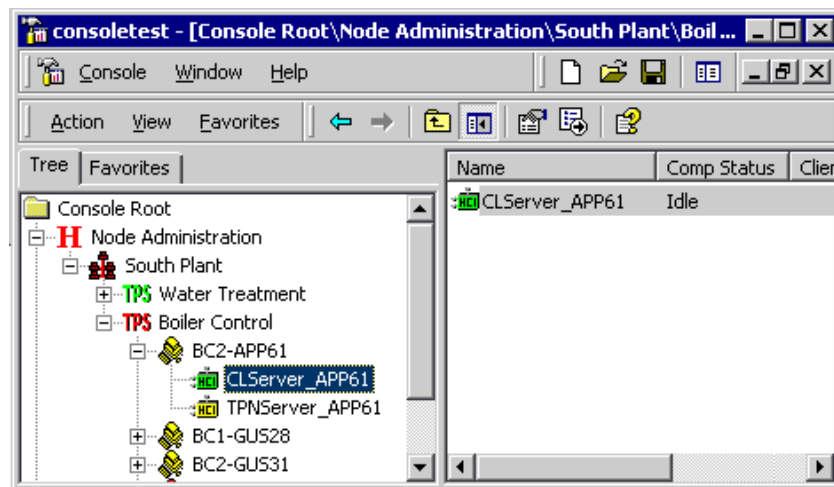
Confirmation of CL Server Startup

Step	Action
4	If the CL Server is in the Stopped state, it can be started up. a) Right click the HCI component in the results pane (this example_uses CLServer_APP61). b) Click All Tasks > Start from the context menu.



CL Server is Started and in Idle Mode

Step	Action
5	Observe the State column in the status frame changed from Stopped to Idle . See step 3 for Status descriptions.



4. Monitoring CL Server Status

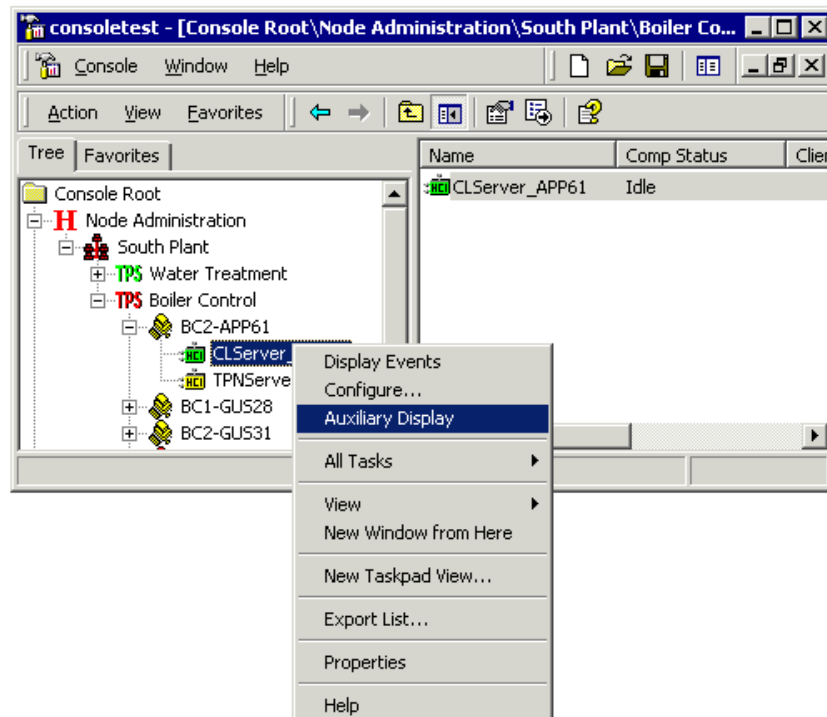
4.5. Invoking the CL Server Auxiliary Status Display

4.5 Invoking the CL Server Auxiliary Status Display

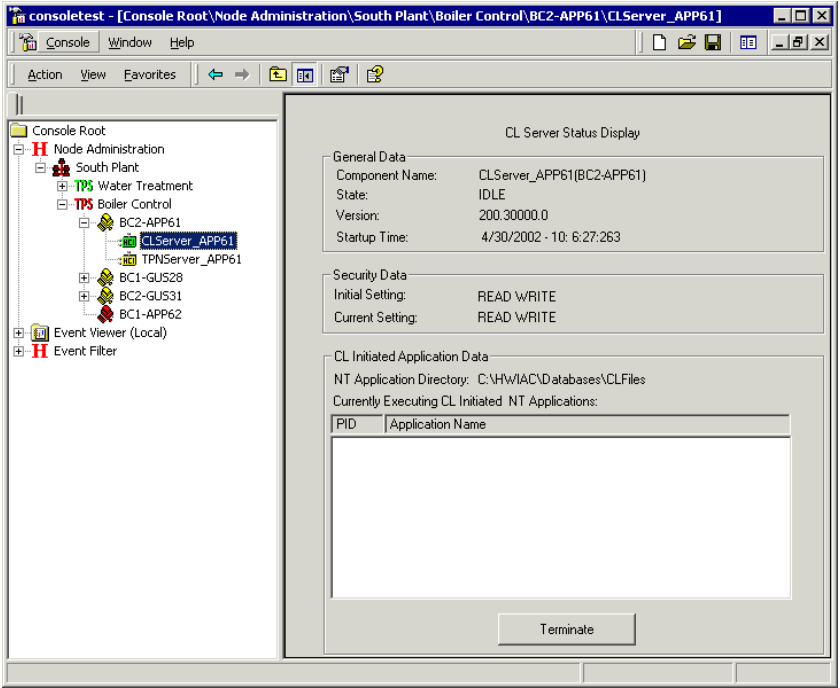
Invoke the Auxiliary Display

The Auxiliary Status Display provides additional information to engineering personnel and is described in more detail in the following sections.

Step	Action
1	Invoke the System Management Display using one of the methods in the procedure <i>Starting-up the System Management Display</i> .
2	Right click the <u>selected</u> CL server. This example uses CLServer_APP61 . A pop-up menu appears with several entries. Note: The Auxiliary Display <i>cannot</i> be invoked in the Stopped or Fail states.



View the CL Server Status Display

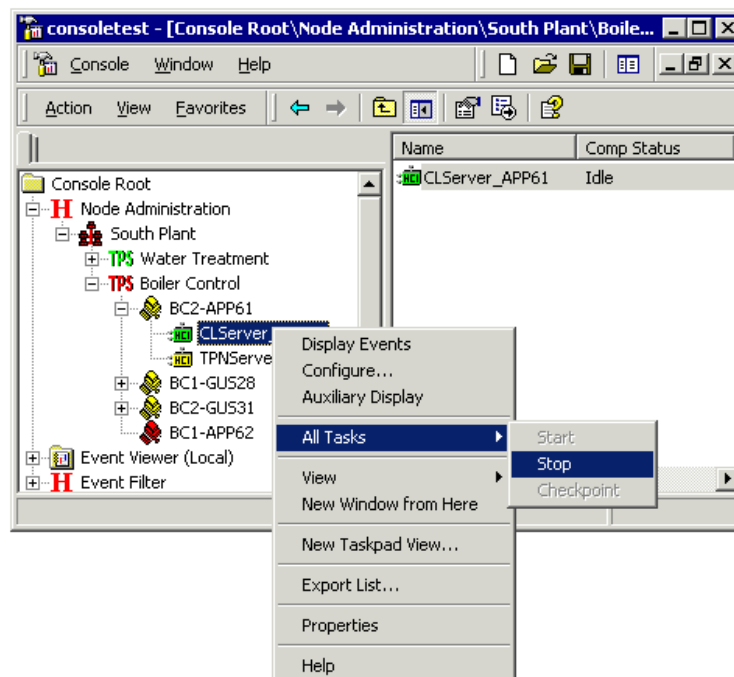
Step	Action
3	Click Auxiliary Display. Result: The CL Server Status Display appears as shown. This is also referred to as the Auxiliary Status Display. Note: See Section 4.7 for a description of the three Data sections contained in the status frame. 
4	To close the Auxiliary Status Display, right-click the CL Server and select Auxiliary Display from the menu.

4.6 Shutting Down the CL Server

Begin the CL Server shutdown

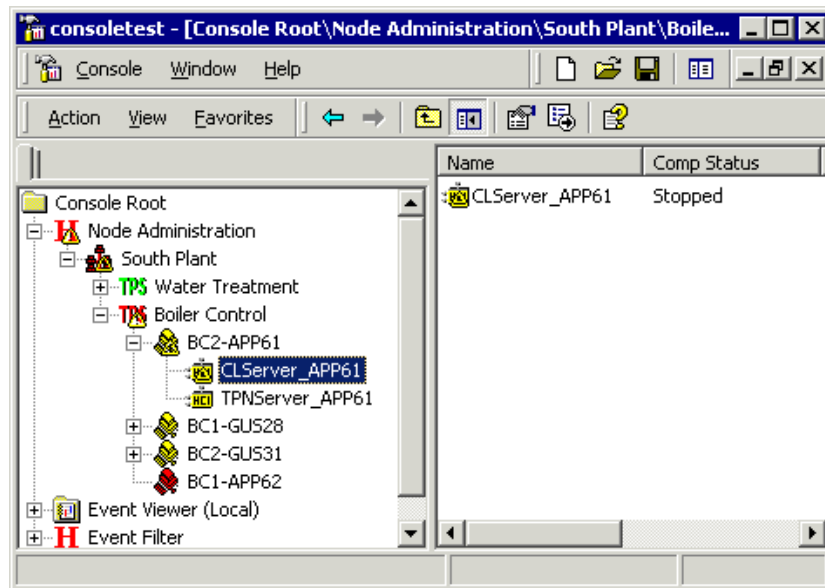
Step	Action
1	Invoke the System Management Display using one of the methods in the procedure <i>Starting-up the System Management Display</i> .
	ATTENTION You must be logged on as the System administrator or as a user that has shutdown permissions to perform the actions in the following steps
	2 Right-click the HCI component that represents the CL Server that you want to shut down and select All Tasks>Stop

Note: If the CL Server is in the Running or Idle state, it can be shut down from the System Management Display.



Shut down the CL Server

Step	Action
3	Observe the CL server transition to a shutdown state. Result: The State of the HCI component CLServer_APP61 transitions to Shutting Down and returns to Stopped or Idle . The CL Server shutdown is complete.



4. Monitoring CL Server Status

4.7. Auxiliary Status Display – CL Server

4.7 Auxiliary Status Display – CL Server

CL Server Auxiliary Status Display organization

The CL Server Auxiliary Status Display is divided into three sections: CL Server General Data, Security Data, and CL-initiated Application Data.

Figure 5 CL Server Status Display

The screenshot shows a window titled "CL Server Status Display". It contains three main sections:

- General Data:**
 - Component Name: CLServer_APP61(BC2-APP61)
 - State: IDLE
 - Version: 200.30000.0
 - Startup Time: 4/30/2002 - 10: 6:27:263
- Security Data:**
 - Initial Setting: READ WRITE
 - Current Setting: READ WRITE
- CL Initiated Application Data:**
 - NT Application Directory: C:\HW\IAC\Databases\CLFiles
 - Currently Executing CL Initiated NT Applications:

PID	Application Name
-----	------------------

At the bottom right of the window is a "Terminate" button.

General Data section

Component Name	The name of the managed component. The host name is shown in parentheses.
State	The State of the currently active CL Server. The Auxiliary Status display is activated only after the component starts; and therefore, only a subset of the possible HCI states is available for viewing. Due to their transitory nature some of these states will never be shown. The State will be one of the following values: FAILED – The server has reached an unrecoverable error and has failed. IDLE – The server is operational, and no CL Initiated Applications are running. INITIALIZING – The server is initializing. RUNNING – The server is operational, and CL Initiated Applications are running. SHUTDOWNCOMPLETE – The server shutdown has completed. SHUTTINGDOWN – The server is shutting down. TEST – The component is performing a self-test. WARNING – The server is running, but there is no connection to the TPS Network.
Version	The version number of the currently active CL Server
Startup Time	The time and date the CL Server was started.

4. Monitoring CL Server Status

4.7. Auxiliary Status Display – CL Server

Security Data section

Initial Setting	The initial setting of the \$XACCESS security.
Current Setting	The current setting of the \$XACCESS security.

CL-Initiated Application Data section

Application Directory	The path to the directory where all the CL-Initiated Applications reside. This directory can contain the actual executable (such as, .exe), or a shortcut (such as, .lnk) to an executable in a different directory.
Currently Executing CL-Initiated Applications	The PIDs (Process Identifier), and the names of the currently executing CL-Initiated Applications
Terminate Button	This button becomes active when you select one or more of the CL-Initiated Applications. When this button is clicked, the selected applications will be terminated.



ATTENTION

If the value is displayed as “?”, then the data are either inaccessible or the CL Server is no longer running.

If the CL Server has failed, the State value will display **FAILED**, and the rest of the values will display “?”.

5. Supporting CL Applications

5.1 Application Directory Summary

General

The information in this subsection is an expansion of the definition of the Application Directory shown on the CL Server Status Display (Auxiliary Status Display). It explains the use of shortcuts, precedence in launching CL-Initiated applications and accessing data using Application Directory.

Using shortcuts



ATTENTION

When a shortcut is created in the Windows Explorer, there is a hidden ".lnk" extension appended to the file name. This extension is not visible in the Windows Explorer. It is visible though, in the old Windows File Manager (winfile.exe). Thus, it is possible to have an executable "xyz.exe" in the Application Directory, and a shortcut to "xyz.exe" in the same directory. The real file name of the shortcut would be "xyz.exe.lnk". The files will appear in the Windows Explorer as having the same name, but are different files.

Precedence in launching CL Initiated applications

When the CL Server receives a request from the TPN Network, to launch a CL-Initiated application, the following order of precedence is used to determine the application to launch:

1. If the file exists, exactly as typed in the command line received from the CL Block, it is launched. For example, if the file name was "xyz.exe", and "xyz.exe" exists in the Application Directory, "xyz.exe" is launched. If the file name was "xyz.exe.lnk" (**Note:** The ".lnk" extension will be hidden in the Windows Explorer), and "xyz.exe.lnk" exists in the Application Directory, the shortcut will be de-referenced, and the de-referenced executable will be launched.
2. If the file does not exist, as typed in the command line received from the CL Block, an ".exe" extension is appended onto the filename. A check is then made to see if the file exists with the ".exe" extension. If it does exist, it is launched. For example, if the file name was "xyz", an ".exe" is appended onto the file name. The file name "xyz.exe" is then checked to see if it exists.

5. Supporting CL Applications

5.1. Application Directory Summary

3. If the file still does not exist, a ".lnk" extension is appended onto the filename. A check is then made to see if the file exists with the ".lnk" extension. If it does exist, the shortcut is de-referenced, and the de-referenced executable is launched. For example, if the file name was "xyz", a ".lnk" is appended onto the file name. The file name "xyz.lnk" is then checked to see if it exists.

Accessing data for CL-Initiated application



ATTENTION

The CL Server, and all CL Initiated Applications run under the TPSApp user account (a member of the Programs global group). If a CL Initiated Application is accessing a data file, make sure that TPSApp has appropriate access to the file and directory where the data file exists. Also, if the Application Directory contains a shortcut to a given CL Initiated Application, make sure that TPSApp account (or Programs global group) has, at minimum, read and execute access to the file and directory where the executable exists

5.2 CL Specific Language Support

Supporting Functions

The following background CL calls are available in the AMCL06_2 set in TPS Network release R433 and later:

- **AMCL06\$Initiate_Task** — Initiates a Windows application capable of receiving background CL events.
- **AMCL06\$Activate_Task** — Sends an activate event to a CL-initiated Windows application.
- **AMCL06\$Terminate_Task** — Sends a terminate event to a CL-initiated Windows application, providing a graceful shutdown of the application.
- **AMCL06\$Execute_Task_With_Wait** — Initiates an application on the Windows-side running the CL Server.

Initiate_Task

The background CL subroutine **AMCL06\$Initiate_Task** is used to initiate a new instance of a Windows application. After the call is made, the background CL suspends execution and enters a wait condition until the Windows application has initiated. The user provides an application name, and a unique application id is returned to the CL block. The CL should store this application id in a CDS parameter or other point.parameter and then complete the execution of the CL block. The application id returned from the subroutine is needed to subsequently activate (**AMCL06\$Activate_Task**) and terminate (**AMCL06\$Terminate_Task**) the application.

Syntax of the **AMCL06\$Initiate_Task** subroutine:

```
SUBROUTINE AMCL06$Initiate_Task

(Ret_Status      : OUT NUMBER; -- Return status of the call
Det_Status       : OUT NUMBER; -- Detailed return status
Appl_ID          : OUT STRING;  -- Application identifier
Cmd_Line         : IN  STRING;  -- Windows application command line
NT_Task_Timeout  : IN  TIME;    -- NT timeout value
Req_Timeout      : IN  TIME)    -- TPN-side timeout value
```

Refer to Appendix I of the CL/AM Reference Manual for more information about the functionality of the call and for definition of the arguments.

Alternate Approach to Launch Windows-Side Hibernating Applications

The CL Server provides one method for initiating side applications from CL blocks. This is done with the subroutine **AMCL06\$Execute_Task_With_Wait**. The approach to launching Windows-side hibernating applications, through the subroutine **AMCL06\$Initiate_Task**, is not supported. If this functionality is desired, an alternate approach to implementing hibernation is described in the following section.

The CL Server is capable of running up to ten applications at a given time. These applications execute from the time the CL Server launches them until they terminate normally or exit with an error. Thus, if a given application is launched numerous times, it runs from start to finish each time. By contrast, an application that uses hibernation behaves in a different manner. It is designed to be launched once, and remain persistent in memory between executions. This offers the following advantages:

- The initialization required by the application is performed only once.
- Data remains persistent between executions.
- While the application is hibernating, the CL Server does not manage it. Thus the hibernating application does not count against the maximum of ten running applications supported by the CL Server.
- The shutdown of the application is performed only once.

Alternate Approach

Hibernation can be programmatically simulated by having a CL Block activate a surrogate application on the Windows-side through the CL Server. The surrogate application can be a simple application that has the ability to launch and communicate with hibernating applications.

Surrogate Application

The surrogate application acts as a middleman between the CL Block and the hibernating application. It receives commands from the CL Block in the form of command line arguments. It then uses Windows events to pass the command to the hibernating application. Its purpose is to signal the hibernating application that a command has been requested by a CL Block. It then suspends its processing until it receives acknowledgement that the hibernating application has completed processing the command. It then exits.

The command line, passed from the CL Block, is of the following form:

<surrogate app> <command> [<hibernating app>] <point name>

Where:

Surrogate app – The name of the surrogate application that is launched through the CL Server. This is the application that the CL Block is requesting the CL Server to run.

Command – One of the following commands:

- **Initiate** – Initiate a new instance of a hibernating application. The name of the hibernating application must be passed with this command.
- **Activate** – Send an activate event to a hibernating application. This event will signal the hibernating application to activate its process algorithm.
- **Terminate** – Send a terminate event to a hibernating application to begin its shutdown logic and exit.

Hibernating app (Optional – only needed with the **Initiate** command) - The name of the hibernating application to be launched by the surrogate application.

Point Name – Passed to insure a level of uniqueness for communications between the surrogate application and the hibernating application. The point name is used to create unique events described in the following section.

Unique Events

Windows Events are used as the mechanism for communication and synchronization between the surrogate application and the hibernating application. By this approach, it is very likely that the user will create one surrogate application, and multiple hibernating applications. Thus, unique events must be used to communicate between the one surrogate application and the many hibernating applications. To make the events unique, the point name is used in the event name. Since the point names are unique on the TPN Network, the following convention will create unique event names on Windows:

CL_<Point Name>_[Activate|Hibernate|Terminate|TermCmplt]

Note: the Point Name is passed to the surrogate application from the CL Block.

The following events must be supported:

- **CL_<Point Name>_Activate** – This event is signalled by the surrogate application when it receives the Activate command from the CL Block. This event will signal the hibernating application to activate its process algorithm.

- **CL_<Point Name>_Hibernate** – This event is signalled by the hibernating application when it enters its hibernating state. It is intended that the surrogate application will be waiting on this event to become signalled after it has processed an **Initiate** command or an **Activate** command. By signalling this event, the hibernating application informs the surrogate application that it has completed its task.
- **CL_<Point Name>_Terminate** – This event is signalled by the surrogate application when it receives the **Terminate** command from the CL Block. This event will signal the hibernating application to begin its shutdown logic and exit.
- **CL_<Point Name>_TermCmplt** – This event is signalled by the hibernating application when it has completed its shutdown logic and is exiting. It is intended that the surrogate application will be waiting on this event to become signalled after it has processed the **Terminate** command.

Hibernating Application

The hibernating application is an event driven application that is launched by the surrogate application. This application is launched by a `CreateProcess()` call done by the surrogate application when it received the **Initiate** command from a CL Block. Upon completing its initialization logic, it enters a hibernating state in which it suspends its process. While in the hibernating state, it waits until one of the following events becomes signaled: **CL_<Point Name>_Activate** event or **CL_<Point Name>_Terminate** event.

When the **CL_<Point Name>_Activate** event becomes signaled, the process algorithm is executed. Upon completing the process algorithm, it signals the **CL_<Point Name>_Hibernate** event to notify the surrogate application that the process algorithm has completed, and it is now hibernating.

When the **CL_<Point Name>_Terminate** event becomes signaled, the shutdown logic is executed. Upon completing the shutdown logic, and prior to exiting, it signals the **CL_<Point Name>_TermCmplt** event to notify the surrogate application that the shutdown has completed, and it is now exiting.

Limitations with Approach

This approach has the following limitations:

- The subroutine **AMCL06\$Get_Hiber_Task_Status** is not supported. Thus, there is not a provided method for getting the current status of the hibernating application.
- The CL Server and all CL initiated applications run under the **TPSApp** account. When the surrogate application does a **CreateProcess** on the hibernating application, the hibernating application may also run under the **TPSApp** account. If the hibernating application runs under the **TPSApp** account, a user will not have access to terminate application with tools like the Task Manager. In cases where a hibernating application hangs, the user may have to re-boot Windows to terminate the application.



ATTENTION

The account the hibernating application runs under it at the discretion of the author of the surrogate application. The author has the option of implementing the **CreateProcess** call to launch a process under a different user.

- The hibernating application is not monitored by the CL Server, and thus will not appear on the CL Server Auxiliary Status Display.

Activate_Task

The background CL subroutine **AMCL06\$Activate_Task** is used to send an activate event to a CL-initiated Windows application. After the call is made, the background CL suspends execution and enters a wait condition until the Windows receives and processes the event. The user provides an application id (obtained from **AMCL06\$Initiate_Task**) and an event string, which is passed to the application.

Syntax of the **AMCL06\$Activate_Task** subroutine:

```
SUBROUTINE AMCL06$Activate_Task

(Ret_Status      : OUT NUMBER;      -- Return status of the call
 Det_Status      : OUT NUMBER;      -- Detailed return status
 Appl_ID         : IN  STRING;       -- Application identifier
 Event_String     : IN  STRING;      -- String passed to application
 NT_Task_Timeout  : IN  TIME;        -- NT timeout value
```

5. Supporting CL Applications

5.2. CL Specific Language Support

```
Req_Timeout      : IN  TIME)          -- TPN-side timeout value
```

Refer to Appendix I of the CL/AM Reference Manual for more information about the functionality of the call and for definition of the arguments.

Terminate_Task

The background CL subroutine **AMCL06\$Terminate_Task** is used to send a terminate event to a CL-initiated Windows application. This provides a graceful shutdown of an application. After the call is made, the background CL suspends execution and enters a wait condition until the application has received and processed the event and terminated its execution. The user provides an application id (obtained from **AMCL06\$Initiate_Task**) and an event string, which is passed to the OpenDDA application.

Syntax of the **AMCL06\$Terminate_Task** subroutine:

```
SUBROUTINE AMCL06$Terminate_Task

(Ret_Status      : OUT NUMBER;        -- Return status of the call
Det_Status       : OUT NUMBER;        -- Detailed return status
Appl_ID          : IN  STRING;        -- Application identifier
Event_String     : IN  STRING;        -- String passed to application
NT_Task_Timeout  : IN  TIME;          -- NT timeout value
Req_Timeout      : IN  TIME)          -- TPN-side timeout value
```

Refer to Appendix I of the CL/AM Reference Manual for more information about the functionality of the call and for definition of the arguments.

Execute_Task_With_Wait

The **AMCL06\$Execute_Task_With_Wait** subroutine is used to initiate an application on the Windows-side running the CL Server. It can only be called from background CL. After the call, the background CL suspends execution and goes into a wait condition until the Windows-side application terminates normally or terminates because an error was detected.

The following is the syntax of the **AMCL06\$Execute_Task_With_Wait** subroutine:

```
SUBROUTINE AMCL06$Execute_Task_With_Wait  
  
  (Ret_Status      : OUT NUMBER; -- Return status of the call  
   Det_Status      : OUT NUMBER; -- Detailed return status  
   Cmd_Line        : IN  STRING;  -- NT-side application command line  
   NT_Task_Timeout : IN  TIME;    -- NT-side timeout value  
   Req_Timeout     : IN  TIME)    -- LCN-side timeout value
```

5. Supporting CL Applications

5.2. CL Specific Language Support

Ret_Status and Det_Status

Ret_Status is the return status of the subroutine call. Det_Status (Detail Status) may contain additional information. Values are:

Value	Ret_Status	Det_Status
0	Application terminated normally	Application exit code (defined in the application program)
1	Subroutine argument error	1 = Invalid NT_Task_Timeout 2 = Invalid Req_Timeout 3 = Invalid Cmd_Line
2	CL timeout while waiting for NT-side application to complete—application aborted	0
3	Unable to communicate with NT-side	0 = No detail status 4 = Unexp'd connect from NT-side 5 = Unexp'd disconnect from NT-side
4	Error getting memory in AM	0
5	LCN/NT-side connection down	0
90	Initialization in progress—attempting connection to NT-side (should only occur during node startup)	0
91	Initialization unsuccessful—unable to acquire internal resources	0
92	Initialization in progress—attempting connection to NT-side (should only occur during node startup)	0
93	Initialization unsuccessful—unable to acquire internal resources	0
500	Internal error	0
501	Internal produce error	0
502	Internal consume error	0

Value	Ret_Status	Det_Status
503	Internal priority error	0
504	Internal message/queue mismatch	0
999	Node not an AMW	0
1001	Application was killed by signal	Signal number
1002	Application name invalid or application not found	NT errno (or 0)
1003	Application timed out (NT-side timeout)—application aborted	0
1004	Miscellaneous internal error	0
1005	An error occurred while changing priority	NT errno
1006	An error occurred while creating the application's process	NT errno
1007	The application did not have any execute permission set	0
1008	An error occurred while executing the application	NT errno
1009	A miscellaneous NT error occurred in the CDS/CL Server	NT errno
1010	The application command line contained an absolute path, which is not allowed	0
1011	An error occurred while setting up the application's environment	0
1012	Function unavailable	This function of AM/CL is not supported in APP nodes.

Cmd_Line

This argument is passed to the NT-side where it is interpreted as an NT command line. It is a string of up to 78 characters containing the application name, not including the pathname, and any application arguments. The application is responsible for interpreting the command line arguments. The application or a link to the application must reside in the directory "c:\HWIAC\Databases\CLFiles." See the "NT Application Directory Summary" section in this document for further information.

Cmd_Line Restrictions

The NT command line can contain multiple arguments, but the first argument must be the NT-side application name. The Application name should not include an explicit directory path. Cmd_Line is a CL local string variable, and is therefore limited to a maximum of 78 characters.

Cmd_Line is not processed for NT shell commands, and therefore does not support items such as:

- NT DOS shell commands such as .bat files
- Multiple commands separated with ";"
- Applications that have User Interfaces.
- Input/output redirection such as: |, <, >, >>

NT_Task_Timeout

This argument passes a time value to the NT-side, which is the maximum time allowed for the application to run before it is aborted and an error status returned. A zero time value disables the NT-side timeout function. Valid time values are in the range from 0 to 24 hours.

Req_Timeout

This argument contains the maximum time allowed for the CL request to wait for an application completion return before an error status is returned. This timer function is performed on the LCN-side. If a timeout occurs, an NT-side application abort is requested. A zero value disables the LCN timeout function and is equivalent to an infinite timeout. Valid time values are in the range from 0 to 24 hours.

Req_Timeout Example

```
PACKAGE
--
BLOCK EXECTASK ( GENERIC; AT BACKGRND )
--
%INCLUDE_SET AMCL06
--
LOCAL STRTAPPL                : STRING
LOCAL RET_STAT, DET_STAT      : NUMBER
LOCAL NT_TMOUT, REQTMOUT      : TIME
--
CALL BKG_CHANGE_PRIORITY( LOW )
SET STRTAPPL = "APP1"
SET NT_TMOUT = 1 MINS
SET REQTMOUT = 2 MINS
--
CALL AMCL06$EXECUTE_TASK_WITH_WAIT (RET_STAT,
&                                DET_STAT,
&                                STRTAPPL,
&                                NT_TMOUT,
&                                REQTMOUT)
--
SEND: "EXIT CODE = ",DET_STAT
--
IF (RET_STAT <> 0.0) THEN
& SEND: "ERROR: STATUS = ", RET_STAT
--
END EXECTASK
- -
END PACKAGE
```

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